

Cybersecurity and Modern Threat Protection

Introduction

Cybersecurity protects systems, networks, and applications from cyberattacks and unauthorized access.

Major Threats

• Phishing attacks • Ransomware • DDoS attacks • SQL Injection • Zero-day vulnerabilities

Real-Time Challenges

1. AI-generated phishing emails. 2. Cloud security vulnerabilities. 3. Insider threats. 4. Data breaches and credential theft. 5. IoT device exploitation.

Real-Time Examples

• Ransomware attacks on hospitals. • Data breaches affecting millions of users. • Banking malware targeting financial systems. • DDoS attacks on gaming servers.

Best Practices

• Multi-factor authentication • Endpoint protection • Security monitoring • Encryption • Regular patching and audits